

State space models I

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University of Chicago

Department of Economics

Mikkel Plagborg-Møller

State space models

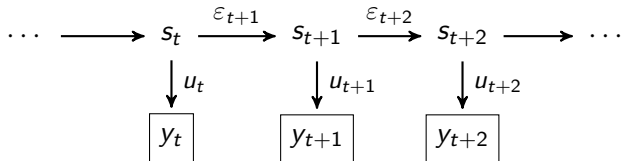
- **State space model** (also known as **hidden Markov model**):

$$y_t = g(s_t, u_t; \theta) \quad (\text{observation equation})$$

$$s_t = h(s_{t-1}, \varepsilon_t; \theta) \quad (\text{transition equation})$$

$$(u'_t, \varepsilon'_t)' \stackrel{i.i.d.}{\sim} F_\theta \quad (\text{shock distribution})$$

- y_t : n -dim. observed data. s_t : m -dim. unobserved **state variables**.
- $g(\cdot)$, $h(\cdot)$, and F known up to parameter vector θ .
- Assume $\{u_t\}$ and $\{\varepsilon_t\}$ are independent (can always redefine s_t).



Linear-Gaussian state space models

- Important special case: **linear-Gaussian state space models**.

$$y_t = c + Hs_t + u_t \quad (\text{observation equation})$$

$$s_t = Fs_{t-1} + \varepsilon_t \quad (\text{transition equation})$$

$$u_t \stackrel{i.i.d.}{\sim} N(0_{n \times 1}, R)$$

$$\varepsilon_t \stackrel{i.i.d.}{\sim} N(0_{m \times 1}, Q) \quad (\text{shock distribution})$$

$$\{\varepsilon_t\} \perp\!\!\!\perp \{u_t\}$$

- Parameters: c, H, F, R, Q . Could themselves be functions of deeper parameters.

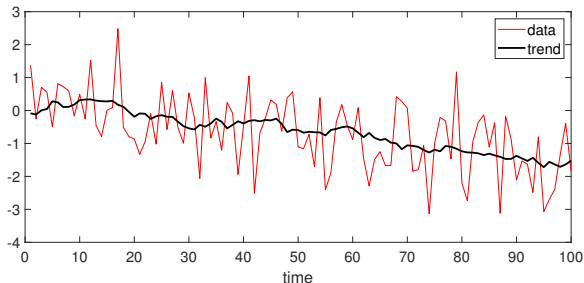
Example: local level model

$$y_t = \mu + s_t + u_t,$$

$$s_t = s_{t-1} + \varepsilon_t,$$

$$(u_t, \varepsilon_t)' \stackrel{i.i.d.}{\sim} N(0_{2 \times 1}, \text{diag}(\sigma_u^2, \sigma_\varepsilon^2)).$$

- Assume σ_ε^2 is small.
- Then s_t is the slow-moving (random walk) *trend* in the data y_t , while u_t is irregular noise.



Example: dynamic factor model

$$\underbrace{y_t}_{n \times 1} = \mu + \lambda \underbrace{s_t}_{1 \times 1} + u_t,$$

$$s_t = \rho s_{t-1} + \varepsilon_t,$$

$$(u'_t, \varepsilon_t)' \stackrel{i.i.d.}{\sim} N(0_{(n+1) \times 1}, \text{diag}(\sigma_1^2, \dots, \sigma_n^2, \sigma_\varepsilon^2)).$$

- **Factor** s_t is a scalar underlying measure of the business cycle that is manifested in the n observables y_t .
- Here all serial correlation and cross-correlation in y_t are assumed to be due to s_t , but that could be relaxed.
- Could generalize to more than one latent factor.

Example: ARMA(p, q)

- Consider the causal ARMA(p, q):

$$\rho(L)(y_t - \mu) = \theta(L)\varepsilon_t, \quad \varepsilon_t \stackrel{i.i.d.}{\sim} N(0, \sigma^2),$$

$$\rho(L) = 1 - \sum_{\ell=1}^p \rho_\ell L^\ell, \quad \theta(L) = 1 + \sum_{\ell=1}^q \theta_\ell L^\ell.$$

- Exercise:** Show that if we set $m \equiv \max\{p, q + 1\}$ and interpret $\rho_\ell = 0$ for $\ell > p$ and $\theta_\ell = 0$ for $\ell > q$, then we can write the ARMA model in state space form:

$$y_t = \mu + (1, \theta_1, \theta_2, \dots, \theta_{m-1})s_t,$$

$$s_t = \begin{pmatrix} \rho_1 & \rho_2 & \rho_3 & \dots & \rho_m \\ 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & & \vdots \\ 0 & 0 & \dots & 1 & 0 \end{pmatrix} s_{t-1} + \begin{pmatrix} \varepsilon_t \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Filtering and likelihood

- Back to general non-linear/non-Gaussian case.
- Given knowledge of θ , how can we learn about current hidden state s_t from data $\mathcal{Y}_t \equiv \{y_\tau\}_{1 \leq \tau \leq t}$ up to time t ?
- Recursive **filtering** algorithm:
 - ▶ Given $p(s_{t-1} \mid \mathcal{Y}_{t-1}, \theta) \dots$
 - ▶ Forecast s : $p(s_t \mid \mathcal{Y}_{t-1}, \theta) = \int \underbrace{p(s_t \mid s_{t-1}, \theta)}_{\text{transition}} p(s_{t-1} \mid \mathcal{Y}_{t-1}, \theta) ds_{t-1}.$
 - ▶ Forecast y : $p(y_t \mid \mathcal{Y}_{t-1}, \theta) = \int \underbrace{p(y_t \mid s_t, \theta)}_{\text{measurement}} p(s_t \mid \mathcal{Y}_{t-1}, \theta) ds_t.$
 - ▶ Update via Bayes' rule:

$$p(s_t \mid \mathcal{Y}_t, \theta) = p(s_t \mid y_t, \mathcal{Y}_{t-1}, \theta) = \frac{p(y_t \mid s_t, \theta) p(s_t \mid \mathcal{Y}_{t-1}, \theta)}{p(y_t \mid \mathcal{Y}_{t-1}, \theta)}.$$

Filtering and likelihood (cont.)

- Notice that the filtering algorithm yields the likelihood as a by-product:

$$p(\mathcal{Y}_T \mid \theta) = \prod_{t=1}^T p(y_t \mid \mathcal{Y}_{t-1}, \theta).$$

- Could in principle estimate state space model by maximum likelihood.
- However, requires:
 - ① We can evaluate transition density $p(s_t \mid s_{t-1}, \theta)$.
 - ② We can evaluate measurement density $p(y_t \mid s_t, \theta)$.
 - ③ We can compute integrals on previous slide (at least numerically).
- Tough in many cases, but we will soon discuss tractable special cases.

Smoothing

- Given knowledge of θ , how can we learn about hidden state s_t at time t from *all* available data $\mathcal{Y}_T$?
- Recursive **smoothing** algorithm:
 - ▶ Given $p(s_{t+1} \mid \mathcal{Y}_T, \theta) \dots$

$$\begin{aligned} p(s_t \mid \mathcal{Y}_T, \theta) &= \int p(s_t \mid s_{t+1}, \mathcal{Y}_T, \theta) p(s_{t+1} \mid \mathcal{Y}_T, \theta) ds_{t+1} \\ &= \int p(s_t \mid s_{t+1}, \mathcal{Y}_t, \theta) p(s_{t+1} \mid \mathcal{Y}_T, \theta) ds_{t+1} \\ &= p(s_t \mid \mathcal{Y}_t, \theta) \int \frac{p(s_{t+1} \mid s_t, \theta)}{p(s_{t+1} \mid \mathcal{Y}_t, \theta)} p(s_{t+1} \mid \mathcal{Y}_T, \theta) ds_{t+1}. \end{aligned}$$

- Initialize using $p(s_T \mid y_T, \theta)$ from filtering algorithm, then run smoothing algorithm backwards in time.
- Use $p(s_t \mid \mathcal{Y}_t, \theta)$ and $p(s_{t+1} \mid \mathcal{Y}_t, \theta)$ from filtering algo.

Linear-Gaussian state space models: filtering/smoothing

- Now let us specialize the algorithms to the case of linear-Gaussian state space models.

$$y_t = c + Hs_t + u_t$$

$$s_t = Fs_{t-1} + \varepsilon_t$$

$$\begin{pmatrix} u_t \\ \varepsilon_t \end{pmatrix} \overset{i.i.d.}{\sim} N \left(\begin{pmatrix} 0_{n \times 1} \\ 0_{m \times 1} \end{pmatrix}, \begin{pmatrix} R & 0_{n \times m} \\ 0_{m \times n} & Q \end{pmatrix} \right)$$

- Clearly, $\{y_t, s_t\}$ is a Gaussian process. Thus, joint distribution fully characterized by conditional means and variances/covariances.
- Filter/smoothing algorithms in this case called **Kalman filter/smoother**. Rely on standard formulas for Gaussian cond'l distr's.

Kalman filter

$$(s_t \mid \mathcal{Y}_{t-1}) \sim N(s_{t|t-1}, P_{t|t-1})$$

$$(y_t \mid \mathcal{Y}_{t-1}) \sim N(y_{t|t-1}, V_{t|t-1})$$

$$(s_t \mid \mathcal{Y}_t) \sim N(s_{t|t}, P_{t|t}),$$

where

$$y_{t|t-1} = c + Hs_{t|t-1}$$

$$V_{t|t-1} = HP_{t|t-1}H' + R$$

$$s_{t|t} = s_{t|t-1} + M_t(y_t - y_{t|t-1})$$

$$P_{t|t} = (I_m - M_tH)P_{t|t-1}$$

$$M_t = P_{t|t-1}H'V_{t|t-1}^{-1}$$

$$s_{t+1|t} = Fs_{t|t}$$

$$P_{t+1|t} = FP_{t|t}F' + Q$$

Kalman likelihood and smoother

- Log likelihood contribution (by-product of Kalman filter):

$$\begin{aligned}\log p(y_t \mid \mathcal{Y}_{t-1}) = & -\frac{1}{2}n \log(2\pi) - \frac{1}{2} \log \det(V_{t|t-1}) \\ & - \frac{1}{2}(y_t - y_{t|t-1})' V_{t|t-1}^{-1} (y_t - y_{t|t-1}).\end{aligned}$$

- Kalman smoother:

$$(s_t \mid \mathcal{Y}_T) \sim N(s_{t|T}, P_{t|T}),$$

where

$$\begin{aligned}s_{t|T} &= s_{t|t} + J_t(s_{t+1|T} - s_{t+1|t}) \\ P_{t|T} &= P_{t|t} + J_t(P_{t+1|T} - P_{t+1|t})J_t' \\ J_t &= P_{t|t}F'P_{t+1|t}^{-1}.\end{aligned}$$

Initialize at $t = T$ using filter.

Kalman filter/smoothers: details

- For derivations, see Hamilton (1994) textbook, ch. 13.2, 13.6.
- In absence of prior info about s_1 , usually initialize

$$s_{1|0} = 0_{m \times 1}, \quad \text{vec}(P_{1|0}) = [I_{m^2} - (F \otimes F)]^{-1} \text{vec}(Q).$$

If s_t is non-stationary, could set $P_{1|0} = \tau^2 I_m$ for large τ^2 . Other options in Durbin & Koopman (2012) book.

- Kalman calculations only involve $m \times m$ and $n \times n$ matrices. Computational complexity increases linearly with T .
- Conditional variances do not depend on the realized data $\mathcal{Y}_T$. Feature of Gaussian distributions.
- There exist numerous alternative implementations of the Kalman algorithms that are more numerically efficient+stable in various applications. See Durbin & Koopman (2012) book.

Application: Okun's law factor model

- Small-scale factor model:

$$\begin{pmatrix} \Delta y_t \\ \Delta r_t \end{pmatrix} = \begin{pmatrix} 1 \\ \lambda \end{pmatrix} f_t + \begin{pmatrix} u_{y,t} \\ u_{r,t} \end{pmatrix}.$$

- Data: log industrial production y_t and unemployment rate r_t .
- Don't directly observe state of the economy f_t . Assume it follows stationary AR(1):

$$f_t = \rho f_{t-1} + \varepsilon_t.$$

- The parameters of the model are related to “Okun's law”.
- Distributional assumptions:

$$(u_{y,t}, u_{r,t}, \varepsilon_t)' \stackrel{i.i.d.}{\sim} N(0_{3 \times 1}, \text{diag}(\sigma_y^2, \sigma_r^2, \sigma_f^2)).$$

Application: Okun's law factor model (cont.)

$$\begin{pmatrix} \Delta y_t \\ \Delta r_t \end{pmatrix} = \begin{pmatrix} 1 \\ \lambda \end{pmatrix} f_t + \begin{pmatrix} u_{y,t} \\ u_{r,t} \end{pmatrix},$$

$$f_t = \rho f_{t-1} + \varepsilon_t,$$

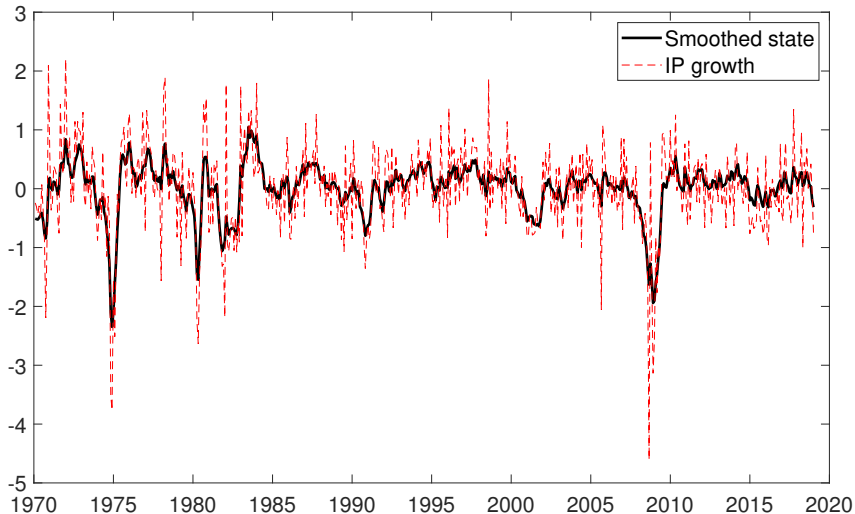
$$(u_{y,t}, u_{r,t}, \varepsilon_t)' \stackrel{i.i.d.}{\sim} N(0_{3 \times 1}, \text{diag}(\sigma_y^2, \sigma_r^2, \sigma_f^2)).$$

- Estimate on monthly U.S. data, seas. adj., 1970:2–2019:1.
- Differenced data is first de-meanned and expressed in percent.
- MLE:

$$\hat{\lambda} = -0.21, \quad \hat{\rho} = 0.82, \quad \hat{\sigma}_f^2 = 0.08, \quad \hat{\sigma}_y^2 = 0.28, \quad \hat{\sigma}_r^2 = 0.02.$$

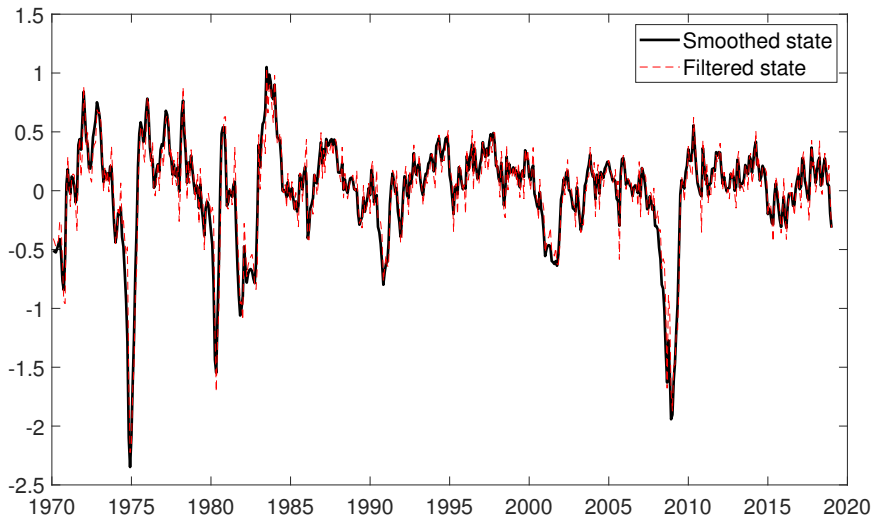
Application: Okun's law factor model (cont.)

Data and smoothed state



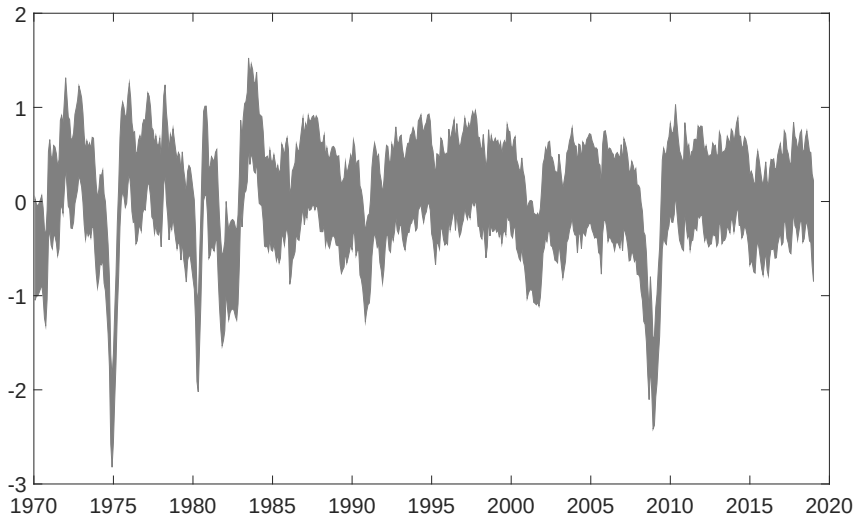
Application: Okun's law factor model (cont.)

Smoothed and filtered state



Application: Okun's law factor model (cont.)

95% uncertainty band for state (from smoothing distribution)



Simulation smoother

- If we want to draw from the smoothing distribution $(s_t \mid \mathcal{Y}_T)$, we can just use the formulas from previous slides and Gaussianity.
- Here's a different approach that is useful much more generally. Called the **simulation smoother** (Durbin & Koopman, 2012 book, ch. 4.9).
- Suppose $\{\mathcal{S}, \mathcal{Y}\}$ are jointly multivariate Gaussian. Here is a method for generating a draw $\tilde{\mathcal{S}} \sim (\mathcal{S} \mid \mathcal{Y})$:
 - ① Simulate $\{\mathcal{S}^*, \mathcal{Y}^*\}$ from the uncond'l joint distribution (no data used).
 - ② Calculate $E(\mathcal{S} \mid \mathcal{Y})$ (using data) and $E(\mathcal{S}^* \mid \mathcal{Y}^*)$ (using simulated data).
 - ③ Set $\tilde{\mathcal{S}} \equiv \mathcal{S}^* + E(\mathcal{S} \mid \mathcal{Y}) - E(\mathcal{S}^* \mid \mathcal{Y}^*)$.
- In state space case, set $\mathcal{S} = \mathcal{S}_T \equiv (s_1, \dots, s_T)$ and $\mathcal{Y} = \mathcal{Y}_T$. Algorithm requires that we simulate from the state space model and run the Kalman mean smoother twice.

Simulation smoother: proof

$$\tilde{\mathcal{S}} \equiv \mathcal{S}^* + E(\mathcal{S} \mid \mathcal{Y}) - E(\mathcal{S}^* \mid \mathcal{Y}^*)$$

- Why does the simulation smoother work?
- Clearly, $\tilde{\mathcal{S}}$ is Gaussian conditional on the data $\mathcal{Y}$. So just need to show that conditional mean and variance are correct.

$$E(\tilde{\mathcal{S}} \mid \mathcal{Y}) = \underbrace{E(\mathcal{S}^* \mid \mathcal{Y})}_{=E(\mathcal{S}^*)} + E(\mathcal{S} \mid \mathcal{Y}) - \underbrace{E[E(\mathcal{S}^* \mid \mathcal{Y}^*) \mid \mathcal{Y}]}_{=E[E(\mathcal{S}^* \mid \mathcal{Y}^*)]=E(\mathcal{S}^*)} = E(\mathcal{S} \mid \mathcal{Y}).$$

$$\begin{aligned}\text{Var}(\tilde{\mathcal{S}} \mid \mathcal{Y}) &= \text{Var}[\mathcal{S}^* - E(\mathcal{S}^* \mid \mathcal{Y}^*) \mid \mathcal{Y}] = \text{Var}[\mathcal{S}^* - E(\mathcal{S}^* \mid \mathcal{Y}^*)] \\ &= \text{Var}(\mathcal{S}^* \mid \mathcal{Y}^*) = \text{Var}(\mathcal{S} \mid \mathcal{Y}),\end{aligned}$$

where the last equality uses that the conditional variance doesn't depend on the realized data (due to Gaussianity).

Kalman filter/smoothing: non-Gaussian case

- Suppose the true disturbances u_t and ε_t are mutually and serially uncorrelated white noise processes, but possibly non-Gaussian.
- Output from Kalman filter/smoothing is then **best linear approx'n**.
- For example:

$$s_{t|t-1} = E^*(s_t \mid 1, \mathcal{Y}_{t-1}), \quad P_{t|t-1} = \text{Var}^*(s_t \mid 1, \mathcal{Y}_{t-1}).$$

- But the filtering/smoothing *distributions* are generally not Gaussian.

Linear state space models: time-varying matrices

- Easy to generalize to the case of *time-varying* coefficient matrices:

$$y_t = c(x_t) + H(x_t)s_t + u_t$$

$$s_t = F(x_t)s_{t-1} + \varepsilon_t$$

$$\begin{pmatrix} u_t \\ \varepsilon_t \end{pmatrix} \Big| \{x_\tau\}_{1 \leq \tau \leq T} \stackrel{\text{indep.}}{\sim} N \left(\begin{pmatrix} 0_{n \times 1} \\ 0_{m \times 1} \end{pmatrix}, \begin{pmatrix} R(x_t) & 0_{n \times m} \\ 0_{m \times n} & Q(x_t) \end{pmatrix} \right).$$

- Here $\{x_t\}$ is a strictly exogenous vector time series.
- Common examples: $c(x_t) = \mu \times t$ (deterministic time trend) or $c(x_t) = \beta x_t$ (exogenous regressors in obs. eq'n).
- Just need to put time subscripts on formulas in Kalman filter/smoothing. See Hamilton (1994) book, ch. 13.8.

Example: time-varying parameter model

- Regression model with time-varying coefficients:

$$y_t = \beta_t' x_t + u_t,$$

$$\beta_t = \beta_{t-1} + \varepsilon_t,$$

$$(u_t, \varepsilon_t)' \stackrel{i.i.d.}{\sim} N(0_{(k+1) \times 1}, \text{diag}(\sigma_u^2, \sigma_1^2, \dots, \sigma_k^2)).$$

- State space model with $H(x_t) = x_t'$ and $s_t = \beta_t$.
- Just one example of a **time-varying parameter (TVP) model**.

Missing data

$$y_t = c(x_t) + H(x_t)s_t + u_t,$$

$$s_t = F(x_t)s_{t-1} + \varepsilon_t,$$

$$\begin{pmatrix} u_t \\ \varepsilon_t \end{pmatrix} \Big| \{x_\tau\}_{1 \leq \tau \leq T} \stackrel{\text{indep.}}{\sim} N \left(\begin{pmatrix} 0_{n \times 1} \\ 0_{m \times 1} \end{pmatrix}, \begin{pmatrix} R(x_t) & 0_{n \times m} \\ 0_{m \times n} & Q(x_t) \end{pmatrix} \right).$$

- A big advantage of the Kalman filter/smoothing approach is that it is straight-forward to optimally handle missing data.
- Let x_t be a vector of indicators for whether the individual time series in y_t are missing at time t . Define $y_{j,t} = 0$ if $x_{j,t} = 1$ (i.e., missing).
- Set $c_j(x_t) = 0$, $H_{j\bullet}(x_t) = 0_{1 \times m}$, $R_{jj}(x_t) = 0$ whenever $x_{j,t} = 1$.
- This simply amounts to not updating s_t based on data from $y_{j,t}$ at time t in Kalman filter. Computationally easy.
- Can obtain $E(y_{j,t}^{\text{true}} \mid \mathcal{Y}_T)$ from Kalman smoother output.

Mixed-frequency data

- A special case of missing data is when we observe time series at different frequencies, e.g., monthly and quarterly.
- Can specify a state space model where latent state vector is defined at highest frequency. Low-frequency data is missing in the observation equation in between low-frequency data points.
- Example:
 - ▶ Have data on monthly industrial production (IP) and quarterly GDP.
 - ▶ State: latent measure of real total output, monthly.
 - ▶ IP enters in obs. eq'n every period.
 - ▶ GDP only enters obs. eq'n in March, June, September, December, and is related to the sum of the past three months' state variables.

Application: NY Fed nowcast

2026:Q2 | 2026:Q1 | 2025:Q4 | 2025:Q3

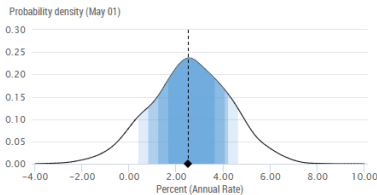
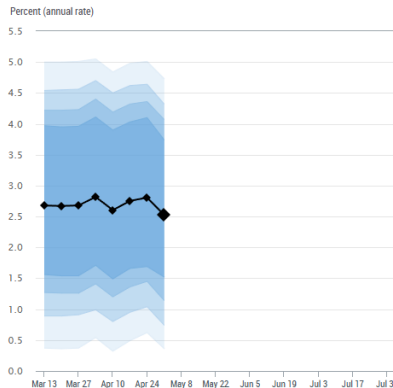
[Reset to latest nowcast](#)

Latest Release: May 1, 2026



New York Fed Staff Nowcast

- ◆ The New York Fed Staff Nowcast 50% Probability 60% Probability 70% Probability 80% Probability
□ Latest GDP Estimate ○ Advance GDP Estimate

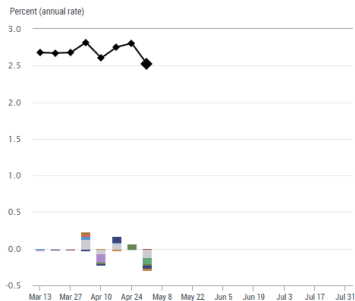


Probability intervals

Release Date	Series	Value
May 1	◆ Nowcast	2.52
	50% Probability	1.52, 3.74
	60% Probability	1.14, 4.07
	70% Probability	0.74, 4.32
	80% Probability	0.36, 4.73
Apr 24	◆ Nowcast	2.81

Accessed on May 6, 2026

Impact of Data Releases



[Expand](#)

Data Flow (May 1, 2026)

Model update	Release Date	Data Series	Actual	Impact	Nowcast GDP Growth
1-May					2.52
	10:01AM May 01	ISM mfg.: Employment index	46.40	-0.01	
	10:01AM May 01	ISM mfg.: Prices index	84.60	0.01	
	10:01AM May 01	ISM mfg.: PMI composite index	52.70	-0.04	
	8:33AM Apr 30	Real gross domestic product	1.99	-0.07	
	8:31AM Apr 30	PCE less food and energy: Chain price index	0.29	0.00	
	8:31AM Apr 30	PCE: Chain price index	0.66	0.00	
	8:31AM Apr 30	Real disposable personal income	-0.06	-0.01	
	8:34AM Apr 30	Real personal consumption expenditures	0.24	-0.02	
	8:30AM Apr 29	Building permits	-166.00	-0.02	
	8:30AM Apr 29	Housing starts	10.77	0.03	
	8:31AM Apr 29	Manufacturers' inventories: Durable goods	0.24	0.00	
	8:31AM Apr 29	Manufacturers' unfilled orders: All industries	0.11	0.00	
	8:31AM Apr 29	Manufacturers' shipments: Durable goods	0.69	0.00	
	8:37AM Apr 29	Total business inventories	0.90	-0.01	
	8:35AM Apr 29	Merchant wholesalers: Total inventories	1.36	-0.01	
	8:31AM Apr 29	Manufacturers' new orders: Durable goods	0.83	-0.01	
	8:30AM Apr 29	Building permits	152.00	0.01	
	8:30AM Apr 29	Housing starts	-3.00	-0.01	
		Data revisions		0.01	
		Parameter revisions		-0.13	

24-Apr

2.81

Regime-switching models

- Consider an AR(1) model with several **regimes**:

$$y_t = c_j + \rho_j y_{t-1} + u_t, \quad u_t \stackrel{\text{indep.}}{\sim} N(0, \sigma_j^2), \quad j = 1, 2, \dots, N.$$

- Define $s_t = j$ if regime j applies at time t .
- Assume s_t evolves according to a **discrete Markov chain** without feedback from $\{y_t\}$:

$$p(s_t = j_2 \mid s_{t-1} = j_1, \{s_\tau\}_{\tau \leq t-2}, \mathcal{Y}_{t-1}) = \underbrace{p(s_t = j_2 \mid s_{t-1} = j_1)}_{\equiv \pi_{j_1 j_2}}.$$

- Matrix of transition probabilities:

$$\Pi \equiv \begin{pmatrix} \pi_{1,1} & \dots & \pi_{1,N} \\ \vdots & \ddots & \vdots \\ \pi_{N,1} & \dots & \pi_{N,N} \end{pmatrix}$$

Regime-switching: general case

- More generally, consider any regime-switching model satisfying

$$p(y_t \mid \mathcal{Y}_{t-1}, \{s_\tau\}_{1 \leq \tau \leq t}, \tilde{\theta}) = p(y_t \mid \mathcal{Y}_{t-1}, s_t, \tilde{\theta}).$$

- State s_t evolves as discrete Markov chain on $\{1, 2, \dots, N\}$ without feedback from $\{y_t\}$. Transition prob matrix Π .
- We observe $\{y_t\}$ but not $\{s_t\}$.
- This is a state space model with a discrete state.
- Would like to estimate $\theta \equiv \{\tilde{\theta}, \Pi\}$ and $P(s_t \mid \mathcal{Y}_T, \theta)$.

Regime-switching: filtering/smoothing

- Filtering/smoothing algo from Hamilton (1994) textbook, ch. 22.4.
- Define vector $\xi_{t|\tau}$ whose j -th element is $P(s_t = j \mid \mathcal{Y}_\tau, \theta)$.
- Define vector η_t whose j -th element is $p(y_t \mid \mathcal{Y}_{t-1}, s_t = j, \theta)$.
- Filter:
 - ▶ $\xi_{t|t} \propto \xi_{t|t-1} \circ \eta_t$, where “ $\circ$ ” denotes element-wise multiplication.
 - ▶ $\xi_{t+1|t} = \Pi' \xi_{t|t}$.
- Likelihood contribution: $p(y_t \mid \mathcal{Y}_{t-1}, \theta) = \xi'_{t|t-1} \eta_t$.
- Smoother: $\xi_{t|\tau} = \xi_{t|t} \circ \{\Pi(\xi_{t+1|\tau} \div \xi_{t+1|t})\}$, where “ $\div$ ” denotes element-wise division.

Application: regime changes in U.S. data

- Sims & Zha (AER 2006) estimate a regime-switching VAR model on monthly U.S. data, 1959–2003.
- 6 variables: commodity price, money stock, fed funds rate, interpolated GDP, cons'n expenditure price level, unempl. rate.
- Restrict the functional form of the time-variation.
- Bayesian approach with Minnesota-type prior.
- Marginal likelihood calculations support a model with *no* regime changes in autoregressive coefficients, but *nine* regimes for volatilities.

Application: regime changes in U.S. data (cont.)

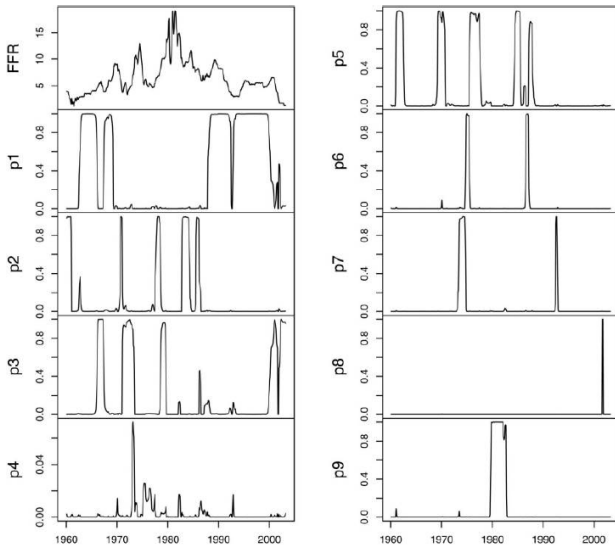


FIGURE 1. NINE-STATE VARIANCES-ONLY PROBABILITIES

Note: The Fed Funds Rate is in the upper left.

Source: Sims & Zha (2006)